B. Tech. Fourth Semester (Information Technology) / (SOE_23 IT-101) Examination

Course Code : 23 IT 1401 **Course Name : Discrete Mathematics and Graph Theory**

Time : 3 Hours] [Max. Marks : 50

Instructions to Candidates :—

(1) Do not write anything on question paper except your exam seat number.

(2) Write the accurate question number in left margin of answer book along with answers.
(3) All questions carry marks as indicated.
(4) Due credit will be given to neatness and adequate dimensions.
(5) Assume suitable data wherever necessary and mention at the beginning of answer.
(6) Diagrams and chemical equations should be given wherever necessary.

1. Solve any **Two** :

(A) Using mathematical induction prove that –

$$1 + 3 + 6 + \cdots \frac{n(n+1)}{2} = \frac{n(n+1)(n+2)}{6}$$

For all $n \in I^*$. 4(CO1)

(B) Write converse, inverse, contrapositive of following statements :

"Rain is sufficient for it to be cloudy." 4(CO1)

(C) Prove that if $A \subseteq B$ then $A \cup (B - A) = B$. 4(CO1)

2. Solve any **Two** :

(A) If A and B are two non-empty sets having n elements in common, then prove that $A \times B$ and $B \times A$ have n^2 elements is common. 4(CO2)

(B) Find the relation R generating the partition $A = \{\{1, 2\}, \{3\}, \{4, 5\}\}$ on set $S = \{1, 2, 3, 4, 5\}$. Also prove that R is on equivalence relation. 4(CO2)

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(C) If R is the set of positive real number then show that a function $f: R^+ \to R^+$ defined as –
$f(x) = 5x^2 - 1 \ \forall \ x \in R^+$ is one-one and onto function. 4(CO2)

3. Solve any **Two** :

(A) Let $G = \{1, -1, i, -i\}$ be a group with respect to multiplication operation $\bullet$, then find the order of each element of group G. 4(CO3)

(B) Prove that set $G = \{1, 2, 3, 4\}$ is finite abelian group with respect to operation "Multiplication modulo 5". 4(CO3)

(C) Let $G = \{1, W, W^2\}$ is set of cube root of unity then show that G is an abelian group under multiplication operation. 4(CO3)

4. Solve any **Two** :

(A) If R is a ring and $a, b \in R$ then show that –
$(a + b)^2 = a^2 + ab + ba + b^2 \neq a^2 + 2ab + b^2$. 4(CO3)

(B) Let $Z = \{0, 1, 2, 3, 4, 5\}$ and $(Z, +_6, X_6)$ is ring with respect to operation $+_6$ and X_6. Then check whether "Z" is integral domain **or** not. 4(CO3)

(C) If R is a ring such that $a^2 = a \ \forall \ a \in R$ then show that –

(i) $a + a = 0 \ \forall \ a \in R$

(ii) $a + b = 0 \Rightarrow a = b \ \forall \ a, b \in R$. 4(CO3)

5. Solve any **Two** :

(A) Let $M = \left\{ \begin{bmatrix} a & b \\ -b & a \end{bmatrix} / a, b \in R \right\}$ is a ring with respect to operation matrix addition and matrix multiplication then show that "M" is a field. 4.5(CO3)

(B) Let $A = \{a, b, c\}$ and $L = \{0, 1\}$.
Let P(A) denotes the power set of A then show that $(P(A), \subseteq)$ and $(L^3, \leq)$ are isomorphic. 4.5(CO3)

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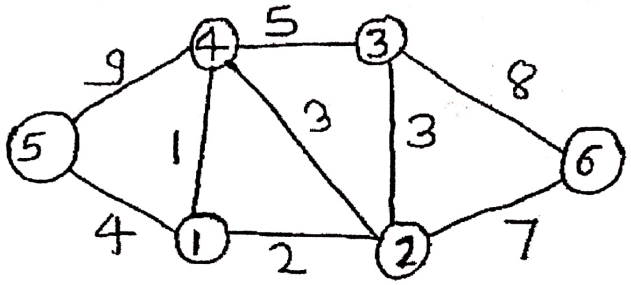
(C) Let $(R, +, \bullet)$ is a ring where –

$R = \{a + b\sqrt{2} / a, b \in Q\}$

Where Q is set of rational numbers then shown that R is field with respect to same operations. 4.5(CO3)

6. Solve any **Two** :

(A) In newly planned town represented by graph, each circle denotes major junction and edges between them represent potential roads with associated construction costs. Design the network such that all junctions are connected with minimum total cost, without forming any cycle.



4.5(CO4)

(B) Draw the diagraphs for the following matrices and hence show that both graphs are isomorphic :

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

4.5(CO4)

(C) Draw the tree for following algebraic structure :

(i) $(a + b) * [(c \div d) - e]$,

(ii) $(a + 5) * [\{(7 * b) + c\} / (9 + d)]$. 4.5(CO4)

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ZDR/2KTF/ET – 241233 To 241274

B. Tech. Fourth Semester (Civil Engineering / ME / EL / EE / ET / CT / IT / CSE / AIDS / AIML / CSD / VLSI / CSEIoT) / (SOE_23 CV - 101) Examination

Course Code : 23 GE 1401 **Course Name : Entrepreneurship Development**

Time : 3 Hours] [Max. Marks : 50

Instructions to Candidates :—

(1) Do not write anything on question paper except your exam seat number. 407155

(2) Write the accurate question number in left margin of answer book along with answers.

(3) All questions are compulsory.

(4) All questions carry marks as indicated.

(5) Due credit will be given to neatness and adequate dimensions.

1. Solve any **Two** of the following :

(A) Explain types of entrepreneurs and provide examples to illustrate their roles in business on society. 6(CO1)

(B) Discuss the difference between Entrepreneur and Intrapreneur. 6(CO1)

(C) Discuss the historical evolution of entrepreneurship, highlighting key shift in the role of entrepreneurs during the Industrial Revolution. 6(CO1)

2. Solve any **Two** of the following :

(A) Classify the difference between Company and sole proprietorship. 6.5(CO2)

(B) What are legal compliances required for incorporating a startup and why are they essential for the Business's success ? 6.5(CO2)

(C) What are the key steps involved in incorporating a partnership firm and why is drafting a partnership deed important ? 6.5(CO2)

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3. Solve any **Two** of the following :

(A) State the works eligible for copyright registration in India. What not ? 6.5(CO3)

(B) Elaborate the rights, duties and obligation of patentee. 6.5(CO3)

(C) Define a trademark and explain its various types in terms of brand protection. 6.5(CO3)

4. Solve any **Two** of the following :

(A) What is the significance of government policies for Small-Scale Enterprises (SSEs) and mention key policies aimed at promoting their growth ? 6(CO4)

(B) Explain the concepts of diversification and joint venture and discuss how these strategies can contribute to the growth of a business. 6(CO4)

(C) What is a business incubator ? Explain its types and significance in the early stage of start ups. 6(CO4)

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B. Tech. Fourth Semester (Information Technology) / (SOE_23 IT - 101) Examination

Course Code : 23 IT 1402 **Course Name : Digital Circuits and Microprocessor**

Time : 3 Hours] [Max. Marks : 50

Instructions to Candidates :—

(1) Do not write anything on question paper except your exam seat number. 404155

(2) Write the accurate question number in left margin of answer book along with answers.

(3) All questions are compulsory.

(4) All questions carry marks as indicated.

1. Solve any **Two** :

(A) Simplify the following expression using Boolean laws :

$A = XY + (XZ)' + XY'Z(XY + Z)$ 4(CO1)

(B) Design 8:1 Multiplexer and implement it using Logic gates. 4(CO4)

(C) Design 3:8 Decoder and implement it using Logic gates. 4(CO4)

2. Solve any **Two** :

(A) Convert the given expression into standard canonical SOP and POS form.

$Y = A + BC$ 4(CO1)

(B) Reduce the expression using K-Map :

$F = \Sigma m(0, 1, 2, 3, 5, 6, 8, 9, 11, 12, 13) + \Sigma d(4, 7, 14, 15)$ and implement using logic gates. 4(CO1)

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(C) Convert S-R flip flop to J-K flip flop. 4(CO1)

3. Solve any **Two** :

(A) Design 3 bit Up-Counter using J-K flip flop. 4(CO4)

(B) Design 3 bit Down Counter using J-K flip flop. 4(CO4)

(C) Explain types of Register with block diagram. 4

4. Solve any **Two** :

(A) Explain the functions of following pins of 8086 microprocessor :

(a) INTA,

(b) $RQ/\overline{GT_0}$ and $RQ/\overline{GT_1}$,

(c) NMI,

(d) TEST. 4(CO2)

(B) Elaborate in short physical memory organization in 8086. 4(CO2)

(C) Draw and discuss the internal block diagram of 8086. 4(CO2)

5. Solve any **Two** :

(A) List and explain the various addressing modes present in 8086 with the help of an example. 4.5(CO2)

(B) Write a program to add a data byte located at offset 0500H in 2000H segment to another data byte available at 0600H in the same segment and store the result at 0700H in the same segment. 4.5(CO2)

(C) What is Assembler ? Explain DW, EQU and ASSUME. 4.5(CO2)

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6. Solve any **Two** :

(A) Explain in detail with the modes of operation of 8255. 4.5(CO3)

(B) Write a program using 8086 to interfaces ADC with 8255. 4.5(CO3)

(C) Design an 8086 based system in minimum mode containing 64 KB of EPROM and 64 KB of RAM. 4.5(CO4)

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ZDR/2KTF/ET–241234 To 241275

B. Tech. Fourth Semester (Civil Engineering / ME / EL / EE / ET / CT / IT / CSE / AIDS / AIML / CSD / VLSI / CSEIoT) / (SOE_23 CV-101) Examination

Course Code : 23 GE 1405 **Course Name : Marathi Language**

Time : 3 Hours] [Max. Marks : 50

Instructions to Candidates :—

(1) Do not write anything on question paper except your exam seat number.

(2) Write the accurate question number in left margin of answer book along with answers.
(3) All questions are compulsory.
(4) All questions carry marks as indicated.

1. **Answer any Two : खालीलपैकी कोणतेही दोन प्रश्न सोडवा. (2×6·5=13) :**

(A) (1) आत्मविश्वास मानव जातीचे सर्वाधिक कल्याण करू शकेल असे लेखक स्वामी विवेकानंद का म्हणतात ? 3

(2) उद्योग दिन कसा व का साजरा करावा असे संत तुकडोजी महाराजांचे मत आहे ? 3.5

(B) (1) स्त्रीने गोसाव्याला कोणती विनंती केली ? का ? 2

(2) गोसावीने बाळाला कसे सांभाळले ? 2

(3) स्त्रीने परत आल्यावर शेजाऱ्यांना कोणती चौकशी केली ? 2.5

(C) खालील सर्व प्रश्नांची उत्तरे एका वाक्यात लिहा :

(1) कोणत्या आरती संग्रहासाठी माहीम भटांचे नाव घेतले जाते ? 1

(2) श्री गोविंद प्रभू चरित्र हा ग्रंथ कोणत्या पंथाशी संबंधित आहे ? 1

(3) ज्योतिबा फुले यांच्या मते शिक्षक आणि विद्यार्थी या दोघांचे लक्ष अधिक प्रमाणात कोणत्या परीक्षेवर असते ? 1

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(4) 'तरुणांनो ! निर्भय बना, शूर बना' या पाठाचे लेखक कोण आहेत ? 1

(5) पगारी नोकऱ्या कोणाला मिळतात असे ज्योतिराव फुले यांचे मत आहे ? 1

(6) जोड्या लावा :

(अ) कोंकराची कथा	(1) जोतिराव फुले
(ब) 'नववर्षाच्या उदया चलावर उद्योगाचे ध्वजारोपण'	(2) अरविंद गोखले
(क) शिक्षण	(३) राष्ट्रसंत तुकडोजी महाराज

1.5

2. **Answer any Two : खालीलपैकी कोणतेही दोन प्रश्न सोडवा. (2×6·5=13) :**

(A) (1) व्यावहारिक अभ्यासक्रमाची गरज का आहे ? 3.0

(2) आजच्या काळात भाषांतराचे कोणते महत्त्व आहे ? 3.5

(B) (1) 'हेतूमूलकता' हे भाषेचे वैशिष्ट्य कसे असते ? 2.0

(2) 'पुनरावृत्ती' हे संदेश मजबूत करण्याचे साधन कसे आहे ? 2.0

(3) परस्परांतील भेद व पर्यायाचे तत्त्व याविषयी थोडक्यात माहिती लिहा. 2.5

(C) Fill in the blanks : रिकाम्या जागी योग्य पर्याय लिहा :

(1) पुनरावृत्ती म्हणजे ——— होय.

(अ) एकच गोष्ट अनेकदा सांगणे.

(ब) एकच गोष्ट फक्त एकदाच सांगणे.

(क) एकच गोष्ट अर्धवट सांगणे. 1

(2) ——— हे प्रभावी असे संवादाचे माध्यम आहे.

(अ) विसंवाद

(ब) भाषा

(क) श्रवण 1

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(3) कार्यालयीन पत्रव्यवहारात ——— भाषेची आवश्यकता असते.

•(अ) मुद्देसूद औपचारिक भाषेची

(ब) अनौपचारिक भाषेची

(क) बोली भाषेची

1

(4) भाषिक संवाद व्यवहाराची मुलतत्त्वे एकूण ——— आहेत.

(अ) 5

(ब) 7

•(क) 9

1

(5) रुढ प्रकारचे 'पुस्तकी' शिक्षण घेतलेला विद्यार्थी ——— एकारलेला असतो.

•(अ) बौद्धिकदृष्ट्या

(ब) मानसिकदृष्ट्या

(क) शारीरिकदृष्ट्या

1

(6) चूक किंवा बरोबर ते लिहा :

(1) सारांश लेखन ही एक आकलन प्रक्रिया आहे.

(2) आधुनिक जगातील पत्रलेखन ही एक प्रमुख कला आहे.

(3) जाहिरात लेखनासाठी ग्राहकांच्या मानसिकतेच्या अभ्यासाची आवश्यकता असते.

1.5

3. **Answer any Two : खालीलपैकी कोणतेही दोन प्रश्न सोडवा. (2×6=12) :**

(A) (1) मराठी भाषेचा मोठेपणा कोणते दृष्टांत देऊन कवी मुकुंदराजांनी सांगितलेला आहे ?

3.0

(2) कावळ्याचे ओरडणे विरहिणीला शुभशकुनाचे कां वाटते ? 3.0

(B) (1) 'दवांचे थेंब' या कवितेतील बालकाच्या मनात कशाचे कुतुहल निर्माण झाले आहे ?

2.0

(2) 'दवांचे थेंब' या कवितेतील बाळ हिमकण पाहून काय विचारते ? 2.0

(3) बाळाला हिमकण कसे दिसले ? 2.0

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(C) Fill in the blanks : रिकाम्या जागी योग्य पर्याय लिहा :

(1) ज्ञानेश्वरी या ग्रंथाचे दुसरे नाव ——— हे आहे.

*(अ) भावार्थदीपिका

(ब) विवेकसिंधू

(क) अमृतानुभव 1

(2) 'ज्ञानदेवे रचिला पाया' असे संत ज्ञानेश्वरांच्या संदर्भात ——— यांनी म्हटले आहे.

(अ) जनाबाई

*(ब) बहिणाबाई

(क) कान्होपात्रा 1

(3) रोज कवीला पडणारा मुख्य प्रश्न ——— आहे.

(अ) बेटीचा

*(ब) रोटीचा

(क) घराचा 1

(4) 'नवकाव्याचे प्रणेते' असा गौरव ——— कवीचा करण्यात येतो.

*(अ) बा.सी. मर्ढेकर

(ब) बालकवी

(क) कुसुमाग्रज 1

(5) प्रत्यक्ष जीवनात कवी जसा जगतो तसा तो ——— आहे.

*(अ) शब्दांत

(ब) शहरात

(क) गावात 1

(6) काऊचा पाय विरहीणी ——— मढवीन म्हणते.

*(अ) सोन्याने

(ब) चांदीने

(क) तांब्याने 1

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4. **Answer any Two : खालीलपैकी कोणतेही दोन प्रश्न सोडवा. (2×6=12) :**

(A) (1) एका क्रिकेटपटूच्या (Cricketer) मुलाखतीसाठी दहा प्रश्नांची प्रश्नावली तयार करा ? 3.0

(2) यशवंतराव चव्हाण अभियांत्रिकी महाविद्यालयात क्रीडा महोत्सव उत्साहात संपन्न झाला या विषयावर बातमी लेखन करा. 3.0

(B) कथा वाचून खालील प्रश्नांची उत्तरे द्या :

*** विश्वासाचा विजय ***

आदित्य हा खूप हुशार आणि मेहनती मुलगा होता, पण त्याला नेहमी वाटायचं की तो इतरांपेक्षा कमी आत्मविश्वासाचा आहे. त्याचे आई-वडील त्याला नेहमी प्रोत्साहन देत, पण त्याला स्वतःवर विश्वास बसत नव्हता. शाळेत एक वक्तृत्व स्पर्धा होती आणि शिक्षकांनी आदित्यचं नाव सुचवलं. आधी त्याला खूप भीती वाटली, पण आईने त्याला समजावलं, "तू फक्त स्वतःवर विश्वास ठेव. प्रयत्न केल्याशिवाय कधी कळणार ?"

आदित्यने तयारीला सुरुवात केली. तो प्रत्येक दिवशी आरशासमोर उभा राहून भाषणाचा सराव करू लागला. सुरुवातीला त्याला अवघड जात होतं, पण हळूहळू त्याचा आत्मविश्वास वाढू लागला. स्पर्धेचा दिवस आला. स्टेजवर जाताना त्याचे हात थोडे थरथरत होते, पण त्याने स्वतःला मनात बजावलं - मी हे करू शकतो !

त्याने आत्मविश्वासाने भाषण दिलं. प्रेक्षक टाळ्या वाजवत होते. निकाल लागला आणि त्याच्या नावाची घोषणा झाली - तो प्रथम आला होता ! त्याच्या डोळ्यात आनंदाश्रू तरळले. त्या दिवसापासून त्याला एक धडा मिळाला - आत्मविश्वास आणि प्रयत्न केले तर कोणतीही गोष्ट अशक्य नाही.

प्रश्न :-

(1) आदित्यला कोणती गोष्ट करायला भीती वाटत होती ? 1

(2) त्याच्या आईने त्याला काय सल्ला दिला ? 1

(3) आदित्याने तयारी कशी केली ? 1

(4) स्पर्धेच्या दिवशी त्याच्या मनात काय विचार आले ? 1

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(5) स्पर्धेच्या निकाल काय लागला ?

(6) या कथेतून तुम्हाला काय बोध मिळतो ते तुमच्या शब्दात लिहा.

(C) खालील सर्व प्रश्नांची उत्तरे लिहा :

(1) पुढील इंग्रजी शब्दासाठी मराठी पारिभाषिक शब्द लिहा.

(अ) Varification

(ब) Backlog

(क) Vacancy

(ड) Administration

(2) खालील शब्दभेद सांगा.

(अ) पाणी व पाणि

(ब) शीर व शिर

(क) दिन व दीन

(ड) आदी व आदि

(3) चूक किंवा बरोबर ते लिहा :

(अ) बातमी लेखन भूतकाळामध्ये केले जाते.

(ब) मुलाखतीचे एकूण दहा प्रकार आहेत.

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B. Tech. Fourth Semester (Information Technology) / (SOE_23 IT-101) Examination

Course Code : 23 IT 1408 **Course Name : Cyber Laws**

Time : 3 Hours] [Max. Marks : 50

Instructions to Candidates :—

(1) Do not write anything on question paper except your exam seat number. 407155

(2) Write the accurate question number in left margin of answer book along with answers.

(3) All questions are compulsory.

(4) All questions carry marks as indicated.

1. Solve any **Two** :

(A) Discuss the role of Certifying Authorities (CAs) under the IT Act, 2000. How do they ensure the credibility of digital transactions in India ? 6(CO1)

(B) Explain the provisions related to the punishment for cybercrimes under the IT Act. Discuss the penalties under Section 43 and Section 66 of the IT Act. 6(CO1)

(C) Describe the legal recognition of digital and electronic signatures under Indian law. How do they facilitate secure electronic transactions and authentication ? 6(CO1)

2. Solve any **Two** :

(A) What is the significance of the TRIPS Agreement in shaping global intellectual property rights ? How has it influenced IP law in developing countries like India ? 6.5(CO2)

(B) Explain the significance of domain names in modern e-commerce. How are they protected under intellectual property law ? 6.5(CO2)

(C) Why is protection of semiconductor chips important under IP laws ? 6.5(CO2)

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3. Solve any **Two** :

(A) What are the civil remedies available for copyright infringement in India ? Discuss the impact of these remedies on the protection of creative works. 6.5(CO3)

(B) Discuss the different civil and criminal remedies available for infringement of intellectual property rights. Provide examples to support your answer. 6.5(CO3)

(C) What are the criteria for determining whether an intellectual property right is enforceable ? Discuss the factors that affect the enforcement of such rights in cyberspace. 6.5(CO3)

4. Solve any **Two** :

(A) Discuss the role and responsibilities of Certifying Authorities under the IT Act, 2000, and analyze how they contribute to the security and trust of electronic transactions. 6(CO4)

(B) Identify the legal remedies available for victims of cybercrime under law and evaluate the effectiveness of current legislation in addressing emerging cyber threats. 6(CO4)

(C) What are the key challenges in enforcing privacy rights in cyberspace ? Discuss how the Right to Privacy is addressed under the Indian Constitution and the IT Act. 6(CO4)

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B. Tech. Fourth Semester (Information Technology) / (SOE_23 IT-101) Examination

Course Name : Computer Networks

Course Code : 23 IT 1404

[Max. Marks : 50

Time : 3 Hours]

Instructions to Candidates :—

(1) Do not write anything on question paper except your exam seat number.

(2) Write the accurate question number in left margin of answer book along with answers.

(3) All questions are compulsory.

(4) All questions carry marks as indicated.

(5) Due credit will be given to neatness and adequate dimensions.

(6) Assume suitable data wherever necessary and mention at the beginning of answer.

(7) Diagrams and chemical equations should be given wherever necessary.

1. Solve any **Two** :

(a) For each of the following four networks, discuss the consequences if a connection fails.

(a) Five devices arranged in a mesh topology.

(b) Five devices arranged in a star topology (not counting the hub).

(c) Five devices arranged in a bus topology.

(d) Five devices arranged in a ring topology. 4(CO1)

(b) What are some of the factors that determine whether a communication system is a LAN or WAN ? Discuss in detail. 4(CO1)

(c) Dialog control and synchronization are two responsibilities of the session layer in the OSI model. Which layer do you think is responsible for these duties in the TCP/IP model ? Explain your answer. 4(CO1)

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2. Solve any **Two** :

(a) Discuss Frequency Division Multiplexing and Assume that a voice channel occupies a bandwidth of 4 kHz. We need to multiplex 10 voice channels with guard bands of 500 Hz using FDM. Calculate the required bandwidth. 4(CO2)

(b) What is the significance of the twisting in twisted-pair cable ? Discuss twisted pair in detail. 4(CO2)

(c) Compare and contrast a circuit-switched network and a packet-switched network. 4(CO2)

3. Solve any **Two** :

(a) A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is 1001. What is the actual bit string transmitted and Suppose the third bit from the left is inverted during transmission. How will receiver detect this error ? 4(CO3)

(b) (1) Apply odd parity hamming code so we can transmit the denary value 9 = 0001001.

(2) Apply even parity hamming code so we can transmit the ASCII character 'G' :

ASCII 'G' = 1000111 4(CO3)

(c) What is the importance of window size in the Sliding Window Protocol ? How does it affect throughput ? Discuss Sliding Window Protocol in detail. 4(CO3)

4. Solve any **Two** :

(a) An organization is granted the block 211.17.180.0/24. The administrator wants to create 32 subnets.

(a) Find the subnet mask.

(b) Find the number of addresses in each subnet.

(c) Find the first and the last address in the first subnet.

(d) Find the first and the last address in the last subnet (subnet 32). 4.5(CO3)

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(b) A token bucket scheme is used for traffic shaping. A new token is put into the bucket every 5 μsec. Each token is good for one short packet, which contains 48 bytes of data. What is the maximum sustainable data rate ? Discuss token bucket in detail. 4.5(CO3)

(c) Demonstrate the working of distance vector routing protocol with a neat diagram. 4.5(CO3)

5. Solve any **Two** :

(a) UDP is a message-oriented protocol. TCP is a byte-oriented protocol. If an application needs to protect the boundaries of its message, which protocol should be used, UDP or TCP. Discuss UDP Protocol in detail. 4.5(CO3)

(b) How is congestion controlled ? Discuss in detail about the congestion control mechanisms used in transport layer. 4.5(CO3)

(c) TCP uses a transition state diagram to handle sending and receiving segments. Discuss in detail about this diagram. How it handles flow and control ? 4.5(CO3)

6. Solve any **Two** :

(a) Discuss the working of Diffie-Hellman Key Exchange algorithm in detail with neat sketch. 4(CO4)

(b) Why is RSA considered secure ? Solve RSA, given two prime numbers $p = 7$; $q = 11$, $e = 13$; $M = 5$. 4(CO4)

(c) What is HTTP protocol used for ? What is the default port number of HTTP protocol ? Discuss HTTP protocol in detail. 4(CO4)

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